C

IB MATHEMATICS SL • UNIT 5.01

Topic 5: Differential & Integral Calculus

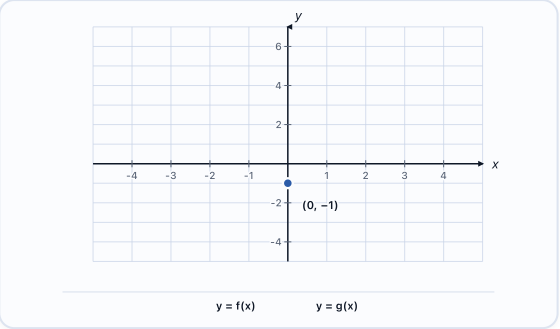
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 5: Differential & Integral Calculus Practice Exercises

1 MCQ

Consider the function $f(x) = x^2 - 5x + 6$. What is the value of $f(2)$?



A 0

B -2

C 4

D 8

2 MCQ

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

A $y = 2x - 5$

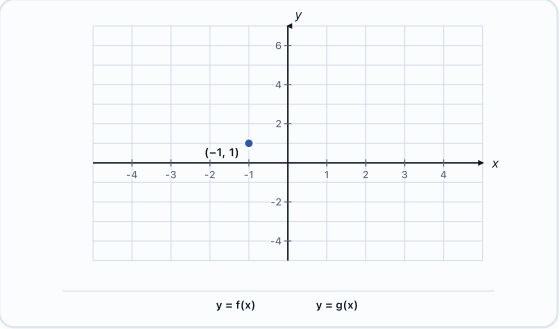
B $y = -\frac{1}{2}x + 5$

C $y = 2x - 8$

D $y = -2x + 11$

3 MCQ

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.



A 1

B 0

C -1

D 2

4 MCQ

Consider the function $f(x) = 3x^2 - 2x + 5$. Find the value of $f'(1)$.

A 4

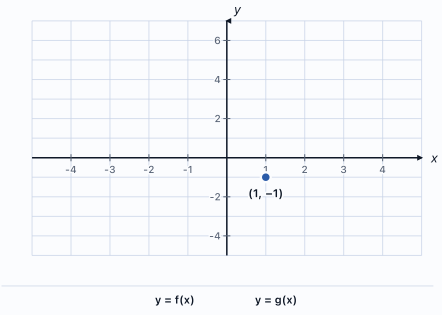
B 5

C 6

D 7

5 MCQ

The graph of the function $f(x) = 2x^2 - 3x + 5$ is shown below. Determine the value of $f'(2)$.



A 7

B 5

C 9

D 2

6 MCQ

Consider the function $f(x) = x^2 - 6x + 5$. Find the x-coordinate of the vertex of the parabola represented by $y = f(x)$.

A 1

B 3

C -1

D -3

7 MCQ

Find $\int x^2 + 4x^3 \, dx$.

A $\frac{1}{3}x^3 + x^4 + C$

B $\frac{1}{4}x^4 + \frac{4}{5}x^5 + C$

C $3x^3 + 16x^4 + C$

D $x^2 + 4x^3 + C$

8 MCQ

Find $\int 5x^2 + 4x \, dx$.

A $\frac{5}{3}x^3 + 2x^2 + C$

B $\frac{5}{4}x^4 + \frac{4}{3}x^3 + C$

C $15x^3 + 8x^2 + C$

D $5x^2 + 4x + C$

9 MCQ

Find $f'(x)$ given $f(x) = x^4 + 4x^2 + 2x$.

A $5x^4 + 12x^2 + 4x$

B $x^4 + 4x^2 + 2x$

C $x^3 + 4x + 2$

D $4x^3 + 8x + 2$

10 MCQ

Find $\int -x^3 - 2\,dx$.

A $-\frac{1}{4}x^4 - 2x + C$

B $-x^3 - 2 + C$

C $-\frac{1}{5}x^5 - x^2 + C$

D $-4x^4 - 2x + C$

11 MCQ

Find $f'(x)$ given $f(x) = 6x^4 - 4x^3 - 2x - 5$.

A $6x^4 - 4x^3 - 2x$

B $30x^4 - 16x^3 - 4x$

C $6x^3 - 4x^2 - 2$

D $24x^3 - 12x^2 - 2$

12 MCQ

Find $\int 3 + 4x^3\,dx$.

A $3 + 4x^3 + C$

B $3x + 16x^4 + C$

C $3x + x^4 + C$

D $\frac{3}{2}x^2 + \frac{4}{5}x^5 + C$

13 MCQ

Calculate $\int 5x^3 + 2x\,dx$.

A $\frac{5}{4}x^4 + x^2 + C$

B $x^5 + \frac{2}{3}x^3 + C$

C $20x^4 + 4x^2 + C$

D $5x^3 + 2x + C$

14 MCQ

Find $\int 6 - 2x^2\,dx$.

A $6x - 6x^3 + C$

B $6 - 2x^2 + C$

C $6x - \frac{2}{3}x^3 + C$

D $3x^2 - \frac{1}{2}x^4 + C$

15 MCQ

Find $\int -4x^2 - 3\,dx$.

A $-12x^3 - 3x + C$

B $-x^4 - \frac{3}{2}x^2 + C$

C $-\frac{4}{3}x^3 - 3x + C$

D $-4x^2 - 3 + C$

16 MCQ

Find $\int -5x + 3x^3\,dx$.

A $-\frac{5}{2}x^2 + \frac{3}{4}x^4 + C$

B $-\frac{5}{3}x^3 + \frac{3}{5}x^5 + C$

C $-10x^2 + 12x^4 + C$

D $-5x + 3x^3 + C$

17 MCQ

Find $\int -2 + 3x\,dx$.

A -2+3x+C

B $-2x + 6x^2 + C$

C $-2x + \frac{3}{2}x^2 + C$

D $-x^2 + x^3 + C$

18 MCQ

Find $\int -2x^2 + 5\,dx$.

A $-\frac{2}{3}x^3 + 5x + C$

B $-\frac{1}{2}x^4 + \frac{5}{2}x^2 + C$

C $-2x^2 + 5 + C$

D $-6x^3 + 5x + C$

19 MCQ

Find $f'(x)$ given $f(x) = 2x^3 + x^2 - 4$.

A $2x^3 + x^2$

B $2x^2 + x$

C $8x^3 + 3x^2$

D $6x^2 + 2x$

20 MCQ

Find $\int x - 4x^3\,dx$.

A $x - 4x^3 + C$

B $2x^2 - 16x^4 + C$

C $\frac{1}{3}x^3 - \frac{4}{5}x^5 + C$

D $\frac{1}{2}x^2 - x^4 + C$

21 MCQ

Calculate $\int -3x^3 - 1\,dx$.

A $-\frac{3}{5}x^5 - \frac{1}{2}x^2 + C$

B $-\frac{3}{4}x^4 - x + C$

C $-3x^3 - 1 + C$

D $-12x^4 - x + C$

22 MCQ

Find $f'(x)$ given $f(x) = 5x^4 + 4x^3 - 5x$.

A $25x^4 + 16x^3 - 10x$

B $5x^3 + 4x^2 - 5$

C $5x^4 + 4x^3 - 5x$

D $20x^3 + 12x^2 - 5$

23 MCQ

Find $f'(x)$ given $f(x) = 4x^4 + 5x^2 + 4x + 1$.

A $4x^4 + 5x^2 + 4x$

B $20x^4 + 15x^2 + 8x$

C $16x^3 + 10x + 4$

D $4x^3 + 5x + 4$

24 MCQ

Find $\int -x - 3x^3\,dx$.

A $-\frac{1}{3}x^3 - \frac{3}{5}x^5 + C$

B $-x - 3x^3 + C$

C $-2x^2 - 12x^4 + C$

D $-\frac{1}{2}x^2 - \frac{3}{4}x^4 + C$

25 MCQ

Find $f'(x)$ given $f(x) = 3x^4 - 3x^2 - 4x$.

A $3x^4 - 3x^2 - 4x$

B $12x^3 - 6x - 4$

C $15x^4 - 9x^2 - 8x$

D $3x^3 - 3x - 4$